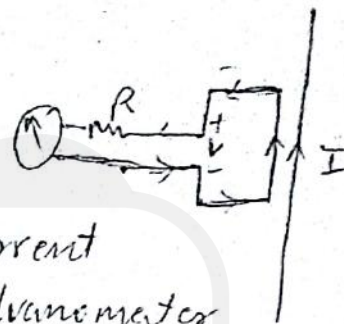


UNIT-3STEADY MAGNETIC FIELD

11

FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION:

The setup shown have a current carrying conductor having current, I and a loop containing a galvanometer and a resistance R . If current



I is decreased to zero, the galvanometer is deflected which mean a current is flowing in the loop. This current is due to an induced voltage, V , in the loop ~~etc~~ by virtue of flux linkage of current carrying conductor and the loop.

Magnetic flux may be defined as the time integral of voltage induced in the loop throughout the interval during which magnetic field is being established (or interval in which the field is decreased to zero.) i.e.

$$\phi = \pm \int_0^t V dt \quad \text{--- (1)}$$

$$\Rightarrow \boxed{V = - \frac{d\phi}{dt}} \quad \text{--- (2)}$$

Unit of flux is Weber (wb) or V-s

If circuit contains N numbers of turns, then

$$\boxed{V = - N \frac{d\phi}{dt}} \quad \text{--- (3)}$$

Maxwell's Eq. from Faraday's law:-

As we know $\phi = \vec{B} \cdot \vec{A} = \int_S \vec{B} \cdot d\vec{S}$

Eq. (3) can be written as, $V = - \frac{d}{dt} \int_S \vec{B} \cdot d\vec{S} \quad \text{--- (4)}$

Also

$$V \Rightarrow \oint \vec{E} \cdot d\vec{l} = - \frac{d}{dt} \oint_S \vec{B} \cdot d\vec{s}$$

Interchanging order of integration & differentiation,

$$\therefore \boxed{\oint \vec{E} \cdot d\vec{l} = - \oint \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}}$$

This is the integral form of Maxwell eq.

Applying Stokes's theorem,

$$\oint_S (\nabla \times \vec{E}) \cdot d\vec{s} = - \oint \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$$

$$\Rightarrow \boxed{\nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}}$$

Magnetic Flux Density ($\vec{B}$) :-

The magnetic flux per unit area through a loop of small area is called Magnetic flux density, $\vec{B}$.

It is a vector quantity and direction is taken normal to the plane of loop when oriented to enclose maximum flux. The unit of $\vec{B}$ is Tesla (T) or Weber per square meter (wb/m^2).

The magnetic flux through any surface is surface integral of normal component of $\vec{B}$, that is

$$\Phi = \int_S \vec{B}_n \cdot d\vec{s} = \int_S \vec{B} \cdot d\vec{s}$$

Magnetic field Intensity and Magnetomotive Force:-

The mag. field intensity or mag. field strength or magnetizing force or magnetic intensity are inter-changeably used and is denoted by $\vec{H}$. It is a vector quantity having direction same as that of $\vec{B}$. The $\vec{H}$ can be given by

$$\vec{H} = \frac{\vec{B}}{\mu} \Rightarrow \vec{B} = \mu \vec{H}$$

where

μ = Permeability of the system.
 $= \mu_0 \mu_r$

As we know that, for a straight long conductor, (3)

$$B = \frac{\mu I}{2\pi r} = \mu H$$

$$\Rightarrow H = \frac{I}{2\pi r}$$

The line integral of $\vec{H}$ is defined as magnetomotive force.

$$F_m = \oint \vec{H} \cdot d\vec{l}$$

For a circular path about wire, with wire at the center, H has the constant value $I/2\pi r$ and thus

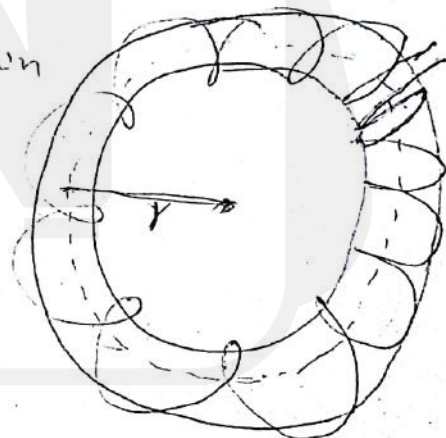
$$F_m = \oint H \cdot dl = I$$

This is known as Ampere's circuital law or Ampere's circuital law.

For a toroidal coil, as shown in fig., consisting of a large no. of closely spaced turns, the mmf is

$$F = NI = \oint H \cdot dl$$

$$\Rightarrow H = \frac{NI}{2\pi r} = \frac{NI}{\ell}$$



where $\ell = 2\pi r$ = length of core.

Permeability (μ) :- Permeability, μ , of the medium can be given by $\mu = \frac{B}{H} = \mu_0 \mu_r$

For free space, $\mu_r = 1$, i.e.

$$\mu = \mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

μ_0, μ_r = Permeability of air & relative perm. of medium respectively.

For air $\mu_r = 1.00000037$. The μ_r can have a value from 1 to several hundred depending upon type of material. But for EM wave theory we can consider it to be fairly constant and in a small range near to 1.

ENERGY STORED IN A MAGNETIC FIELD:-

The amount of work done in establishing a current in a circuit against emf induced by increasing magnetic flux is stored in the field as energy. This energy can be determined by considering the example of toroidal coil. The Magnetic field strength for toroidal coil is given by,

$$H = \frac{nI}{l}$$

n = no. of turns
 l = mean length of path thro core of the coil

The back voltage induced in the coil is,

$$V = -n \frac{d\phi}{dt}$$

$$V = -nA \frac{dB}{dt}$$

$$[\because \phi = BA]$$

A = cross sectional area of core

The work done in establishing current I in the coil is,

$$W = - \int_0^{t_1} V I dt$$

$$= - \int_0^{t_1} \frac{nI}{l} A \frac{d(\mu n)}{dt} dt$$

$$= \int_0^{\mu_1} \mu n A dn$$

$$\left[\begin{array}{l} \text{At } t=0, \mu=0 \\ t=t_1, \mu=\mu_1 \end{array} \right]$$

$$= \frac{\mu A}{2} \left[\mu_1^2 \right]$$

This is the energy stored in the field and $\frac{\mu A}{2}$ is the volume of the coil. Hence, Energy density or Energy stored per unit volume can be given by,

$$w' = \frac{1}{2} \mu H^2 = \frac{\text{Energy stored}}{\text{Vol}}$$

(5)

It states that "line integral of magnetic field ($\vec{H}$) about any closed path is exactly equal to the direct current enclosed by that path."

or
The M.M.F. around a closed path is equal to the current enclosed by that path!
This can be given in integral form as,

$$\oint \vec{H} \cdot d\vec{l} = \vec{I}$$

As we know that $\vec{I} = \int_S \vec{J} \cdot d\vec{S}$

$$\Rightarrow \oint \vec{H} \cdot d\vec{l} = \int_S \vec{J} \cdot d\vec{S}$$

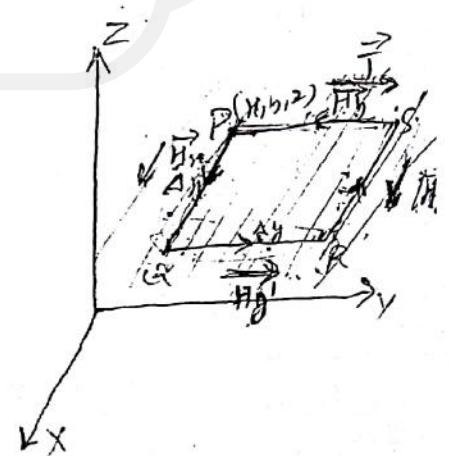
Applying Stokes's theorem on L.H.S, we get

$$\oint_S (\nabla \times \vec{H}) \cdot d\vec{S} = \int_S \vec{J} \cdot d\vec{S}$$

$\Rightarrow \nabla \times \vec{H} = \vec{J}$ $\rightarrow$ This is differential form of Ampere's work law.

$$\Rightarrow \nabla \times \vec{B} = \mu \vec{J}$$

To prove it, consider a loop inside a conducting medium as shown in fig. Let the current density be $\vec{J}$ and component of $\vec{J}$ are $\vec{J}_x, \vec{J}_y$ & $\vec{J}_z$



Now, the magnetomotive force around the closed path PQRS may be obtained by the sum of MMF along four sides of the rectangle.

Let,

$\vec{H}_x$ = Average value of $\vec{H}$ over path PQ.

$\vec{H}_y$ = " " " $\vec{H}$ " " PS.

Now MMF from P to Q = $\vec{H}_x \Delta x$ — (1) [$\because \int_P^Q \vec{H} \cdot d\vec{l} = \vec{H}_x \Delta x$]

MMF " S to P = $-\vec{H}_y \Delta y$ — (2)

Now, $\vec{H}_x'$ & $\vec{H}_y'$ can be given as, ~~etc~~ Even sign shows opposite direction. H & dl -

$$\vec{H}_y' = \vec{H}_y + \frac{\partial H_y}{\partial x} \Delta x$$

$$\vec{H}_x' = \vec{H}_x + \frac{\partial H_x}{\partial y} \Delta y$$

Now, MMF from Q to R = $\vec{H}_y' \Delta y = (\vec{H}_y + \frac{\partial H_y}{\partial x} \Delta x) \Delta y$ — (3)

Also MMF from R to S = $-\vec{H}_x' \Delta x = -(\vec{H}_x + \frac{\partial H_x}{\partial y} \Delta y) \Delta x$ — (4)

Adding eq. (1) to eq. (4), we get

MMF around PQRS = current enclosed by PQRS
[Ampere's work law]

$$\Rightarrow \vec{H}_x \Delta x - \vec{H}_y \Delta y + \vec{H}_y \Delta y + \frac{\partial H_y}{\partial x} \Delta x \Delta y - \vec{H}_x \Delta x - \frac{\partial H_x}{\partial y} \Delta x \Delta y$$

$$= I_z = J_z \Delta x \Delta y \quad [\because I = J \cdot d\vec{s}]$$

$$\Rightarrow \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \Delta x \Delta y = J_z \Delta x \Delta y$$

$$\Rightarrow \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) = J_z \quad \text{--- (5)}$$

Similarly, for elemental Area in y-z plane & z-x plane respectively, we can write,

$$\left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) = J_x \quad \& \quad \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) = J_y \quad \text{--- (6)}$$

Now, adding eq. (5), (6) & (7) vectorially, we get

$$\left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) \hat{x} + \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) \hat{y} + \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \hat{z} = \vec{J}_x \hat{x} + \vec{J}_y \hat{y} + \vec{J}_z \hat{z}$$

$$\Rightarrow \boxed{\nabla \times \vec{H} = \vec{J}} \rightarrow \text{Differential form of Ampere's work law.}$$

AMPERE'S LAW FOR A CURRENT ELEMENT & VOLUME DISTRIBUTION OF CURRENT :-

When $\vec{H}$ due to a closed current carrying conductor is to be determined, it is convenient to consider total $\vec{H}$ as sum contributions from elemental lengths $d\vec{l}$ of the circuit, each carrying current I . This quantity $I d\vec{l}$ is known as current element. It is a vector quantity having direction of current.

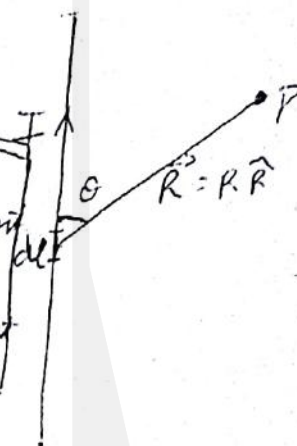
The value of $\vec{H}$ due to $I d\vec{l}$ can be given as.

$$d\vec{H} = \frac{I d\vec{l} \sin \theta}{4\pi R^2}$$

This can also be given as

$$d\vec{H} = \frac{I (d\vec{l} \times \hat{R})}{4\pi R^2}$$

where θ = angle b/w I & line joining the point and $I d\vec{l}$.
 R = Distance of current element & point P.



where $\hat{R}$ = unit vector along $\vec{R}$.

The total field can be given by

$$\vec{H} = \oint d\vec{H} = \oint \frac{I (d\vec{l} \times \hat{R})}{4\pi R^2} \rightarrow \text{This is Ampere's law for current element (or Biot-Savart law).}$$

As we know, current are not necessarily confined to filamentary path, it is useful to generalize Ampere's law to include a current density $\vec{J}$ flowing within a volume V . Under these conditions, $\vec{H}$ can be expressed as,

$$\vec{H}(\vec{r}) = \frac{1}{4\pi} \int \frac{\vec{J}(\vec{r}') \times \hat{R}}{R^2} dV'$$

$$\begin{aligned} \therefore I &= J \cdot dA \\ \Rightarrow I d\vec{l} &= J \cdot d\vec{A} d\vec{l} \\ \Rightarrow I d\vec{l} &= \vec{J} dV \end{aligned}$$

This is the volume distribution of current.

Also

$$\vec{B} = \oint \frac{\mu I (d\vec{l} \times \hat{R})}{4\pi R^2} = \oint \frac{\mu (\vec{J} \times \hat{R})}{4\pi R^2} dV$$

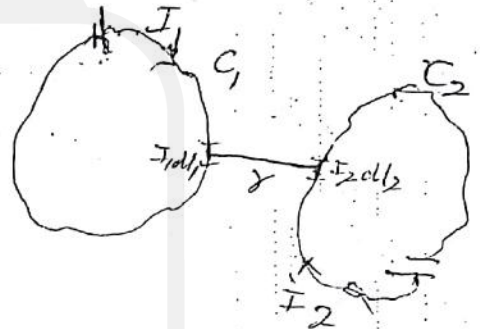
AMPERE'S FORCE LAW:-

According to this, The force, F b/w two parallel conductor is carrying current is proportional to the product of currents in the conductors and inversely proportional to the square of distance between them. i.e.

$$F \propto \frac{I_1 I_2}{r^2}$$

The force b/w two conductors as shown in fig. can be given by

$$F = I_1 (\vec{L} \times \vec{B})$$



$$F_{12} = \oint_C dF_{12} = \oint_{C_1} I_1 (dl_1 \times \vec{B}_2)$$

This is Force on conductor 1 due to 2.
As we know,

$$\vec{B} = \oint \frac{\mu_0 I}{4\pi R^2} d\vec{l} \times \vec{R}$$

$$\Rightarrow \vec{B}_2 = \oint_{C_2} \frac{\mu_0 I_2 (dl_2 \times \vec{R})}{4\pi R^2}$$

$$\Rightarrow F_{12} = \oint_{C_1} \frac{\mu_0 I_1 I_2 (dl_1 \times dl_2 \times \vec{R})}{4\pi R^2} = \oint_{C_1} I_1 (dl_1 \times \oint_{C_2} \frac{\mu_0 I_2 (dl_2 \times \vec{R})}{4\pi R^2})$$

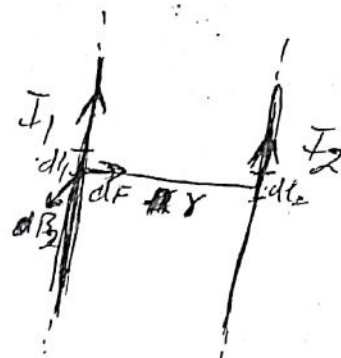
As I_1 & I_2 are constant,

$$\Rightarrow F_{12} = \frac{\mu_0 I_1 I_2}{4\pi} \oint_{C_1} dl_1 \times \oint_{C_2} \frac{dl_2 \times \vec{R}}{R^2}$$

$$\Rightarrow \boxed{F_{12} = \frac{\mu_0 I_1 I_2}{4\pi} \oint_{C_1} \oint_{C_2} \frac{(dl_1 \times dl_2 \times \vec{R})}{R^2}} \rightarrow \text{This is Ampere's Force law.}$$

Force b/w Two Long parallel wires:-

Let us consider two long parallel wires of infinite length, carrying current I_1 & I_2 and separated by a distance r as shown in fig.



The current I_2 will produce a magnetic induction of $\vec{B}_2$ at the position of current I_1 . The force $d\vec{F}_{\text{act}}$ on element dl_1 can be given by,

$$d\vec{F} = I_1 (d\vec{l}_1 \times \vec{B}_2)$$

$$|d\vec{F}| = I_1 dl_1 \frac{\mu I_2}{2\pi r} \quad [\text{as } \theta = 90^\circ]$$

$$= \frac{\mu I_1 I_2}{2\pi r} dl_1$$

$$\Rightarrow \boxed{\frac{dF}{dl} = \frac{\mu I_1 I_2}{2\pi r}}$$

This is the force per unit length on b/w two long infinite current carrying conductors.

MAGNETIC VECTOR POTENTIAL:-

In the electric field, potential was a scalar function of $\vec{r}$ and $\vec{E}$ was obtained from space derivative of V . Similarly, in case of magnetic field it would be desirable to set up a magnetic potential, the space derivative of which gives $\vec{B}$ or $\vec{H}$.

Corresponding to point charges in Electrostatics the source of magnetic field are current elements. Hence, mag. potential would depend on these elements. As Electric potential depends on scalar quantity (point charge) it is also a scalar quantity. In the same manner, Mag. Potential depends upon current elements, thus,

It should be a vector quantity. Hence, it is also known as Mag. Vector Potential. The direction of it is same as that of current element or simply the direction of current.

The strength of Mag. Vector Potential should be directly proportional to $I dl$ and inversely proportional to R as space derivative of it should give magnetic field. $\Rightarrow dA \propto \frac{I dl}{R}$ — (1)

Mag. Vector Potential is denoted by A .

Also, it should be possible to obtain $\vec{B}$ or $\vec{H}$ from space derivative of vector $\vec{A}$. There are two possible space derivative operations on a vector quantity but curl is the only possible space-derivative as divergence results into a scalar quantity but $\vec{B}$ & $\vec{H}$ both are vectors.

Hence

$$\vec{B} = \nabla \times \vec{A} \text{ or } \vec{H} = \nabla \times \vec{A}$$

But in recent year, the former has achieved a wider acceptance, accordingly,

$$\vec{B} = \nabla \times \vec{A} \quad - (2)$$

From eq. (1), $d\vec{A} = \kappa \left(\frac{I dl}{R} \right)$

$$= \frac{\mu I dl}{4\pi R}$$

$\kappa = \text{constant of proportionality}$

$$\Rightarrow \boxed{\vec{A} = \oint \frac{\mu I dl}{4\pi R}}$$

As $I dl = I dV$

$$\Rightarrow \boxed{\vec{A} = \oint \frac{\mu I dV}{4\pi R}}$$



Alternative derivation of $\vec{A}$:-

From Ampere's law for current element, we have

$$\vec{B} = \int \frac{\mu_0 J d\vec{r} \times \vec{R}}{4\pi R^2} \quad \text{--- (1)}$$

In this integrand can be treated like.

$$\frac{J d\vec{r} \times \vec{R}}{R^2} = J d\vec{r} \times \nabla \frac{1}{R} \quad \text{--- (2)} \quad \left[\because \nabla \frac{1}{R} = -\frac{\vec{R}}{R^2} \right]$$

Now, from vector identity,

$$\nabla \times (f \vec{V}) = f (\nabla \times \vec{V}) + \nabla f \times \vec{V} \quad \text{--- (3)}$$

Comparing

putting, $f = \frac{1}{R}$ and $\vec{V} = J d\vec{r}$, we get

$$\nabla \times \frac{J d\vec{r}}{R} = \frac{1}{R} (\nabla \times J d\vec{r}) + \nabla \frac{1}{R} \times J d\vec{r}$$

But J is constant hence $\nabla \times J = 0$

$$\Rightarrow \nabla \times \frac{J d\vec{r}}{R} = \nabla \frac{1}{R} \times J d\vec{r} \quad \text{--- (4)}$$

From eq. (2), we can write.

$$\frac{J d\vec{r} \times \vec{R}}{R^2} = \nabla \frac{1}{R} \times J d\vec{r} \quad \text{--- (5)} \quad \left[\because \vec{A} \times \vec{B} = -\vec{B} \times \vec{A} \right]$$

Comparing eq. (4) & (5), we get

$$\frac{J d\vec{r} \times \vec{R}}{R^2} = \nabla \times \frac{J d\vec{r}}{R} \quad \text{--- (6)}$$

Using eq. (6), eq. (1) can be written as

$$\vec{B} = \oint \frac{\mu_0}{4\pi} \left(\nabla \times \frac{J d\vec{r}}{R} \right) dV$$

Inter changing integral & curl, we get

$$\vec{B} = \nabla \times \oint \frac{\mu_0 J dV}{4\pi R} = \nabla \times \vec{A}$$

where

$$\vec{A} = \oint \frac{\mu_0 J dV}{4\pi R}$$

Thus $\vec{B} = \nabla \times \vec{A}$

BOUNDARY CONDITIONS BETWEEN TWO MAGNETIC MATERIALS:-

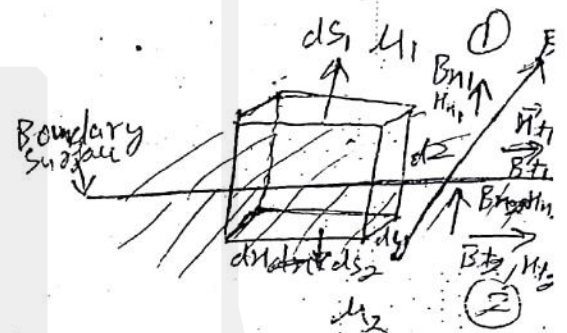
Boundary conditions b/w two magnetic materials states that -

(I) Normal component of $\vec{B}$ is continuous ^{across} the boundary surface. i.e. $B_{n1} = B_{n2}$.

(II) Tangential component of $\vec{H}$ is continuous across the boundary provided no current is flowing through the boundary. i.e. $H_{t1} = H_{t2}$ provided $I = 0$ through boundary.

(I) Normal $B_{n1} = B_{n2}$:-

Let us consider a block of volume $dv = dx dy dz$ such that half of the block is in one material & half is in another medium.



As we know that net magnetic flux from a closed surface is zero i.e.

$$\phi = \int_S \vec{B} \cdot d\vec{s} = 0 \quad \text{--- (1)}$$

Now, The block has six surfaces having three pairs of areas.

$\Delta x \Delta y$, $\Delta y \Delta z$ & $\Delta z \Delta x$

If we assume $\Delta z \rightarrow 0$, then only one pair will be left having area $\Delta x \Delta y$. One of these two faces is just above the boundary i.e. completely in medium ① & other is just below the boundary surface i.e. in medium ②.

B_{n1}, B_{n2} = Normal comp. of $\vec{B}$ in medium ① & ②
 H_{t1}, H_{t2} = Tangential comp. of $\vec{H}$ in ① & ②
 μ_1, μ_2 = Permeability of medium ① & ②
 B_{t1}, B_{t2} = Tangential comp. of $\vec{B}$ in ① & ②
 H_{n1}, H_{n2} = Normal comp. of $\vec{H}$ in ① & ②



Now, eq. ① can be written as,

$$\int_{S_1} \vec{B} \cdot d\vec{S} + \int_{S_2} \vec{B} \cdot d\vec{S} = 0$$

$$\Rightarrow \vec{B} \cdot d\vec{S}_1 + \vec{B} \cdot d\vec{S}_2 = 0$$

The component of $\vec{B}$ in direction of $d\vec{S}_1$ & $d\vec{S}_2$ are B_{n1} & B_{n2} respectively and also from fig. $d\vec{S}_2 = -d\vec{S}_1$,

$$\Rightarrow B_{n1} dS_1 - B_{n2} dS_1 = 0$$

$$\Rightarrow \boxed{B_{n1} = B_{n2}}$$

Also $B_{n1} = \mu_1 H_{n1}$ & $B_{n2} = \mu_2 H_{n2}$

$$\Rightarrow \mu_1 H_{n1} = \mu_2 H_{n2} \quad \text{and } \mu_1 \neq \mu_2$$

$$\Rightarrow H_{n1} \neq H_{n2}$$

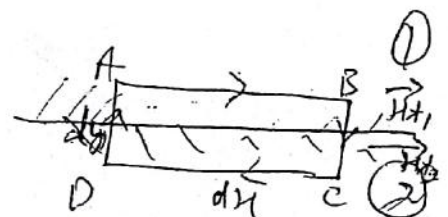
Hence, we can say that Normal component of $\vec{B}$ is continuous across the boundary and Normal component of $\vec{H}$ is discontinuous across the boundary.

④ $H_{t1} = H_{t2}$ Provided $I = 0$:-

consider a closed loop ABED as shown in fig.

Now we know that, from Ampere's Law,

$$\oint \vec{H} \cdot d\vec{l} = I$$



$$\Rightarrow \int_{AB} \vec{H} \cdot d\vec{l} + \int_{BC} \vec{H} \cdot d\vec{l} + \int_{CD} \vec{H} \cdot d\vec{l} + \int_{DA} \vec{H} \cdot d\vec{l} = I \rightarrow \left[\text{Current enclosed by ABCD} \right]$$

Assuming $dy \rightarrow 0$, we have only two terms left

$$\int_{AB} \vec{H} \cdot d\vec{l} + \int_{CD} \vec{H} \cdot d\vec{l} = I \rightarrow \left[\text{Current enclosed is the only current flowing through boundary} \right]$$

$$\Rightarrow H_1 \cdot dl_1 + H_2 \cdot dl_2 = I$$

Now Component of H in the direction of dl is H_1 & H_2 . Also

$$dl_1 = dx \quad \& \quad dl_2 = -dx$$

$$\Rightarrow H_1 dx - H_2 dx = I$$

[Due to opposite direction of H_2 & dx]

$$\Rightarrow H_1 - H_2 = \frac{I}{dx} = K \rightarrow \left[\text{Linear current density} \right]$$

If current through the boundary is zero i.e. $I = 0$ [where $I = \text{current flowing through boundary}$]

Then

$$H_1 - H_2 = 0$$

$$\Rightarrow \boxed{H_1 = H_2}$$

$$\text{Also } \mu_1 = \frac{B_1}{H_1} \quad \& \quad \mu_2 = \frac{B_2}{H_2}$$

$\Rightarrow$

$$\frac{B_1}{\mu_1} = \frac{B_2}{\mu_2}$$

But

$$\mu_1 \neq \mu_2 \Rightarrow$$

$$\boxed{B_1 \neq B_2}$$

Hence, Tangential component of $\vec{H}$ is continuous across the boundary provided no current is flowing through the boundary.
and Tangential component of $\vec{B}$ is discontinuous across the boundary.

EQUATION OF CONTINUITY FOR TIME VARYING FIELDS:-

Since current is simply 'charge in motion', the total current flowing out of some volume must be equal to the rate of decrease of charge within the volume. That is, it can be shown as,

$$I = \oint_S \mathbf{J} \cdot d\mathbf{S} = -\frac{dQ}{dt} \quad \left[\begin{array}{l} \text{-ve sign shows rate of} \\ \text{decrease} \end{array} \right]$$

Now, $Q = \int_V \rho \, dv$

$$\Rightarrow \oint_S \mathbf{J} \cdot d\mathbf{S} = -\frac{d}{dt} \int_V \rho \, dv$$

$$\Rightarrow \oint_S \mathbf{J} \cdot d\mathbf{S} = -\int_V \frac{\partial \rho}{\partial t} \, dv$$

Applying divergence theorem on L.H.S., we get

$$\int_V (\nabla \cdot \mathbf{J}) \, dv = -\int_V \frac{\partial \rho}{\partial t} \, dv$$

$$\Rightarrow \boxed{\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}} \quad \text{This is the time varying form of eq. of continuity.}$$

INCONSISTENCY OF AMPERE'S LAW:-

Ampere's law is consistent for steady currents but it is inconsistent with time varying equation of continuity. This is due to the fact that Ampere's law doesn't involve displacement current & hence can't hold good in case of capacitors. Hence a modification in ampere's law is desired to make it consistent for time varying fields also.

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Maxwell's eq. of continuity for time varying fields (16)

can be given as,

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}$$

Also, from Gauss law,

$$\nabla \cdot \mathbf{D} = \rho$$

$$\Rightarrow \nabla \cdot \mathbf{J} = -\frac{\partial}{\partial t} \nabla \cdot \mathbf{D}$$

$$= -\nabla \cdot \frac{\partial \mathbf{D}}{\partial t}$$

$$\Rightarrow \nabla \cdot \mathbf{J} + \nabla \cdot \frac{\partial \mathbf{D}}{\partial t} = 0$$

$$\Rightarrow \nabla \cdot \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) = 0$$

Maxwell reasoned that this total current density $(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t})$ should replace $\mathbf{J}$ in Ampere's law which gives,

$$\boxed{\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}}$$

This is the modified form of Ampere's law and also one of the Maxwell's equations.

